

B. Sc. (Hons) Semester V Open Book Examination 2021 – 22**PHYSICS****Paper No. BPT-505 : Electromagnetic Theory****Time : 4 hours 30 minutes****Full Marks : 70****Instructions :**

- (i) The Question Paper contains 08 questions out of which you are required to answer any 04 questions. The question paper is of 70 Marks with each question carrying $17 \cdot 5$ Marks.
- (ii) The total duration of the examination will be 4:30 Hours (Four Hours and Thirty Minutes), which includes the time for downloading the question paper from the portal, writing the answers by hand and uploading the hand-written answer sheets on the portal.
- (iii) For the students with benchmark disability as per Persons with Disability Act, the total duration of examination shall be 6 Hours (Six Hours) to complete the examination process, which includes the time for downloading the question paper from the portal, writing the answers by hand and uploading the hand-written answer sheets on the portal.
- (iv) Answers should be hand-written on plain white A4 size paper using black or blue pen. Each question can be answered in up to 350 words on 3 (Three) plain A4 size paper (only one side to be used).
- (v) Answers to each question should start from a fresh page. All pages are required to be numbered. You should write your Course Name, Semester, Examination Roll Number, Paper Code, Paper Title, Date and Time of Examination on the first sheet used for answers.

- Q.1.**
1. For a uniform magnetic field $\vec{B}$, does $\vec{A} = \frac{1}{2}(\vec{r} \times \vec{B})$ qualify for vector potential. Comment on the uniqueness of $\vec{B}$. 4
 2. A square loop of wire with sides of length a lies on a table at a distance d from a very long wire carrying a current I . One pulls out the loop directly away from the wire at a speed v , what e.m.f. is generated? What is the direction of the current flow? $4\frac{1}{2}$
 3. Two uniform infinite sheets with charge densities $-\sigma$ and $-\sigma$ intersect at right angle. Find the magnitude and direction of the electric field everywhere and sketch the field lines. 4
 4. The instantaneous electric field of an e.m. wave is given by $\vec{E} = \hat{a}_x E_{xo} \cos(\omega t - \beta z + \phi_1) + \hat{a}_y E_{yo} \cos(\omega t - \beta z + \phi_2)$. If $\phi_1 = \phi_2 + \pi/2$, $\phi_2 = 0$ and $E_{xo} = E_{yo} = E_0$, determine the polarization and sense of rotation of the wave. 5
- Q.2.**
1. State the fundamental theorem of divergences. What is its geometric interpretation? Evaluate $\int (\vec{\nabla} \times \vec{V}) \cdot d\vec{a}$ for a closed surface of your choice. 7
 2. Show that Maxwell's correction does not change anything to magnetostatics. A FM radio signal amplified to the peak voltage of one hundred volts used to charge a par-

allel plate capacitor of spacing one centimetre. What can you say about the magnetic field generated around a loop with area 10 cm^2 , and the detection of displacement current? $10\frac{1}{2}$

- Q.3.** 1. The electric field in free space is given by $\vec{E} = 50 \cos(10^8 t + \beta x) \hat{a}_y \text{ V/m}$.
 (i) Find the direction of wave propagation.
 (ii) Calculate β and the time it takes to travel a distance of $\lambda/2$.
 (iii) Sketch the wave at $t = 0, T/4$ and $T/2$. $8\frac{1}{2}$
2. What is total reflection? Explain what happens if incident angle θ_i is greater than critical angle θ_c . An electromagnetic wave from an underwater source with perpendicular polarization is incident on a water-air interface at $\theta_i = 20^\circ$. Using $\varepsilon_r = 81$, $\mu_r = 1$ for freshwater, find out critical angle θ_c . **9**
- Q.4.** 1. Show that a rectangular waveguide does not support TM_{01} or TM_{10} modes. **9**
2. Calculate the dimensions of an air-filled rectangular waveguide for which the cut-off frequencies for TM_{11} and TE_{03} modes are both equal to 12 GHz. At 8 GHz, determine whether the dominant mode will propagate or evanescent in the waveguide. $8\frac{1}{2}$
- Q.5.** 1. State the Poynting theorem. Show that the time average Poynting vector is $\frac{1}{2} \text{Re}[\vec{E}_s \times \vec{H}_s^*]$, where $*$ indicate complex conjugate of magnetic field $\vec{H}_s$. **10**
2. In a nonmagnetic medium, $\vec{E} = 4 \sin(2\pi \times 10^7 t - 0.8x) \hat{a}_z$, find the time average power carried by the wave and total power crossing 100 cm^2 of a plane $2x + y = 5$. $7\frac{1}{2}$
- Q.6.** 1. Find the reflection and transmission coefficients of obliquely incident electromagnetic wave when wave electric field is perpendicular to the plane of incidence. **12**
2. A plane electromagnetic wave normally incident on the boundary $z = 0$ between two lossless dielectric medium with $\eta_2 > \eta_1$. Obtain the z -values at which maximum values of $|E|$ occur in the medium 1, where E_1 is the total electric field in medium 1. $5\frac{1}{2}$
- Q.7.** Consider a uniformly distributed charge Q on a disc of radius a .
1. Use elementary methods based on azimuthal symmetry of the charge distribution to find potential at any point on the axis of symmetry. $4\frac{1}{2}$
2. With the aid of (a), find an expression of the potential at any point r ($r > a$) as an expansion in angular harmonics (Hermite polynomials). **10**
3. Determine the extrema points of first three Hermite polynomials. **3**
- Q.8.** 1. Derive Ampere's law in matter starting from Ampere's law in free space including the displacement current term. Why is the polarization current different from the bound current? $5\frac{1}{2}$

2. Explain the physical difference between the Lorentz and Coulomb's gauge in terms of applicability. Check if the following potentials qualify for Coulomb's or Lorentz gauge : **12**

- $V(\vec{r}, t) = 0, \vec{A}(\vec{r}, t) = -\frac{1}{4\pi\epsilon_0} \frac{q^2 t}{r^2} \hat{r}$
- $V(\vec{r}, t) = 0, \vec{A}(\vec{r}, t) = A_0 \sin(kx - \omega t) \hat{a}_y$

ExternalPortals&AcademicResources:

- **ExaminationPortal:** <https://bhu-exam.vercel.app/>
- **StudentResourceCenter:** <https://quashan-me.vercel.app/>
- **AcademicPortfolio:** <https://me-aryan.vercel.app>